

NITHUSIKAN THASAVARAN

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PROFILE

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Computer Engineering graduate with hands-on experience in **AI/ML engineering** and **full-stack software development**, gained through academic projects, a research internship, and industry work. Built and deployed **LLM-based systems**, **ML pipelines**, and **cloud-hosted APIs** across NLP, computer vision, and cybersecurity domains. Eager to grow further in applied AI/ML engineering while contributing meaningfully from day one.

EDUCATION

University of Peradeniya, Sri Lanka

BScEng(Hons) specializing in Computer Engineering

2021 March – 2025 August

3.50 / 4.00 CGPA

Jaffna Hindu College, Sri Lanka

G.C.E Advanced Level | 3A | District Rank 46 | Island Rank 562

2011 – 2019

Z-Score: 2.1369

EXPERIENCE

Nanthi Ventures Inc.

Software Engineer

Feb 2026 – April 2026

Colombo, Sri Lanka

- Shipped full-stack features on a production **Accounting SaaS** — refactoring **Django** APIs and **PostgreSQL** flows, overhauling billing pipelines via **SendGrid**, and delivering responsive **React + TypeScript** UI in agile sprint cycles with PR-based code reviews.
- Collaborated with product leadership to scope a **FinTech ride-hailing platform** with a digital wallet, translating business requirements into API specifications and Payment Acquirer integration plans.

LIRNEasia

Data Science Intern

July 2024 – January 2025

Colombo, Sri Lanka

- NLP Research:** Led an NLP research stream modelling tweet virality across a **multilingual corpus of ~100K tweets** (English, Tamil, Sinhala) — benchmarking **TF-IDF**, **Word2Vec**, **FastText**, and **VADER** across **10+ controlled experiments**, improving ensemble model precision-recall through iterative feature selection.
- ML Pipeline:** Designed and built a production-grade **ML pipeline** automating text preprocessing, feature engineering, model training, and evaluation — reducing iteration time by **40%**; maintained full experiment reproducibility via **MLflow** run tracking and **DVC** dataset versioning.
- ETL & Data Engineering:** Engineered a scalable **data ingestion pipeline** using **Scrapy**, **Selenium**, and **BeautifulSoup** to collect and transform high-volume social media data for downstream ML workflows.
- Graph Analysis:** Performed large-scale hashtag network analysis on **100K+ node graphs** using **Louvain** and **Spectral Clustering** (NetworkX, Neo4j), delivering interactive **Plotly** dashboards for stakeholder reporting.

TECHNICAL SKILLS

LLMs & Gen AI: Llama-3, Gemini, Claude API, OpenAI, HuggingFace Transformers, PEFT, LoRA, Quantisation, Unsloth

RAG & Retrieval: LangChain, ChromaDB, FAISS, Embedding Models (BERT, FastText, Word2Vec), Reranking, Semantic Search

Prompt Engineering: Structured Templates, System Prompts, Few-Shot Design, Output Guardrails, Variation Testing

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, YOLOv8, spaCy, NLTK

MLOps: MLflow (tracking, registry), DVC (versioning), Docker, CI/CD, Git

Data & Pipelines: PySpark, Pandas, NumPy, Scrapy, Selenium, ETL Design, SQL (MySQL, PostgreSQL), MongoDB

Cloud: AWS (EC2, ECR, S3, SageMaker, Lambda, IoT Core)

APIs & Dev: FastAPI, Django, Spring Boot (Java), React.js, TypeScript, RESTful APIs

PROJECTS

1. Multi-Model LLM Customer Support System | Individual | 2024 Apr – Present



- Built a production-grade **multi-model LLM system** integrating **Llama-3**, **Gemini**, and **Claude** under a unified **FastAPI** backend with **model routing logic**, **RAG-based retrieval**, and **prompt-engineered guardrails** — deployed end-to-end on **AWS (EC2, S3)**.

* **LLM Fine-Tuning:** Fine-tuned a **4-bit quantized Llama-3** using **PEFT + Unsloth** on a domain-specific Q&A dataset; all runs, hyperparameters, and artefacts tracked via **MLflow**.

- * **RAG Pipeline:** Built a hybrid retrieval system with **LangChain + ChromaDB + FAISS** — implementing history-aware retrieval, semantic reranking, and follow-up question resolution.
- * **Prompt Engineering:** Designed structured prompt templates with system prompts, few-shot examples, and output guardrails; ran systematic variation experiments to optimise accuracy and response tone.
- * **Model Evaluation:** Benchmarked Llama-3, Gemini, and Claude on domain Q&A accuracy, latency, and cost — results drove the final model routing strategy.
- **Tech Stack:** Python, Llama-3, Gemini, Claude API, LangChain, ChromaDB, FAISS, PyTorch, PEFT, FastAPI, MLflow, AWS.

2. Cross-Layer Cyberattack Detection Framework | Group | 2025 Mar – 2025 Aug

- Final-year project building a detection framework that correlates **network traffic** and **application-layer logs** across **12 million flow records** to identify known and zero-day cyberattacks.
 - * **Data Pipeline:** Designed distributed ingestion and feature engineering using **PySpark** on 12M+ records — handling high-dimensionality, class imbalance, and large-scale train/test splits.
 - * **Known Attack Detection:** Trained **XGBoost and Random Forest** multiclass classifiers for **20+ attack types**, achieving **92–97% detection accuracy**; all experiments tracked via **MLflow**.
 - * **Anomaly Detection:** Applied **semi-supervised learning** with threshold-based traffic separation for unknown (zero-day) attack identification — a practical self-learning detection loop.
- **Tech Stack:** Python, PySpark, Scikit-learn, XGBoost, MLflow, Pandas, NumPy, Matplotlib, Seaborn.

3. Automated Hydroponics IoT System | Group | 2023 Nov – 2024 Jan

- IoT system with real-time sensor monitoring, automated condition control, and a live data ingestion pipeline from devices to the cloud — demonstrating end-to-end system integration across hardware, backend, and frontend.
 - * Integrated IoT sensors via **AWS IoT Core (MQTT QoS 1)**; built the full **Spring Boot (Java)** backend with **JWT auth** and **MySQL** schema, and the **React.js** frontend from Figma specs — deployed on **AWS EC2**.
- **Tech Stack:** Spring Boot (Java), React.js, AWS IoT Core, AWS EC2, MySQL.

4. Vehicle Number Plate Recognition System | Individual | 2024

- Real-time **ANPR system** built for the Sri Lankan context — a two-stage deep learning pipeline handling vehicle detection, plate localisation, character extraction, and domain-specific OCR correction.
 - * Fine-tuned two **YOLOv8** models on local data: one for vehicle detection (including three-wheelers absent from standard benchmarks) and one for number plate localisation tailored to Sri Lankan plate formats.
 - * Extracted characters via **EasyOCR** and improved accuracy using rule-based post-processing correction dictionaries (e.g. S→5, I→1), significantly reducing character-level error rate on local plates.
- **Tech Stack:** Python, YOLOv8, EasyOCR, OpenCV, PyTorch.

5. Engineering Education Unit | Group | 2023

- Actively production-deployed web platform serving as the official EEU site at the University of Peradeniya — built full-stack and hosted on the department server, currently live for staff and learners.
 - * Developed **Spring Boot** backend APIs (search, core MVP endpoints) and implemented **authentication & authorisation** using **JWT + Spring Security**; later upgraded to **Google OAuth** for production.
 - * Built the **React.js** frontend from **Figma** design specs with responsive, cross-device layouts; contributed across the full stack in a team-based Git workflow.
- **Tech Stack:** Spring Boot (Java), React.js, MySQL, Spring Security, JWT, Figma.

CERTIFICATIONS

- Supervised Machine Learning: Regression & Classification – Stanford University (Coursera) 
- Advanced Learning Algorithms – Stanford University (Coursera) 
- Introduction to Java – edX  — Advanced Java: Streams, Lambda, Collections & Generics 
- Introduction to Linux – Linux Foundation 

ACHIEVEMENTS

- ACES Hackfest 2023 – **Winners** of Sustainable Development Category (AI/ML Solution)
- ACES Coders v10.0 – **Finalist** in Competitive Programming Competition

REFERENCES

Prof. Roshan G. Ragel, Dept. of Computer Engineering, University of Peradeniya, roshanr@eng.pdn.ac.lk

Dr. Isuru Nawinne, Dept. of Computer Engineering, University of Peradeniya, isurunawinne@eng.pdn.ac.lk

Eng. Merl Chandana, Research Manager & Team Lead: DAP Team, LIRNEasia, merl@lirneasia.net